

Θ THETA DESK

An autonomous options desk that prices **volatility**, not direction.

18

deterministic risk gates
— LLMs cannot loosen a
single one

4

LLM roles that argue,
veto and size — Claude
+ open models via
Featherless

100%

of decisions replay bit-
for-bit from a hash-
chained journal

4

books running live: real
+ no-gates + no-hedge
+ naive baseline

THE THESIS

Over five days, price prediction is a **coin flip.**
The volatility risk premium is **measurable.**

What everyone builds

«LLM reads headlines → buys calls.»
A directional bet over 5 days: ~50% win rate, minus spread, minus theta.
Negative expected value with extra steps.

What THETA DESK does

Compares ATM implied volatility (Alpaca chain, free feed) against 20-day realized volatility — a deterministic VRP score. Rich vol → sell iron condors, harvest theta.
Cheap vol → buy convexity, micro-size.

Day one proof

The plan said «sell condors». The live signal said RV 13.9% > IV 11.6% — premium was cheap. The agent **went against our own plan** and bought puts instead. The system follows the signal, not the narrative.

Regime map: rich → condors · neutral → wider, half size · cheap → micro long vega · QQQ fallback

ARCHITECTURE — ONE TICK, EVERY 15 MINUTES OF THE SESSION

LLMs decide **whether it's wise.**

Code decides **whether it's allowed.**



Management, every tick

35% target on credit · +60% target on debit · realization policy on idle days · NFP de-risk locks winners $\geq +15\%$ the day before drawdown · halt, never panic flatten

Sizing is earned

Risk budgets grow only from **realized** gains: worst-case 6% → 8%, long-premium sleeve 1.5% → 2.5%. The agent earns the right to take risk

Before any order, the **whole book** is repriced.

Every open leg plus the candidate, over a $\pm 20\%$ underlying-price grid at the judging horizon (deadline + 14 days — nothing expires while judges look), under a base and a stressed-vol scenario, each leg priced off its own underlying's spot. If the worst grid P&L breaches budget — the order is never sent, and the refusal is journaled with the full grid.

Why it matters

This is a client-side implementation of the same worst-case principle as **Alpaca's universal spread rule** for options margin — applied one step earlier in the pipeline. We refuse orders before the broker ever could.

Refusals are the product

An agent that can decline to trade is the point. Every refusal is a first-class journal entry: which gate, why, at what worst-case number. The dashboard shows them next to the fills.

Tested like an engine

58 unit + property tests: condor worst case = one wing; hedge converts the tail, not the plateau; random books never pass beyond budget; missing spot raises instead of silently mispricing.

TRUST, MECHANICALLY

Every number **regenerates**. Every decision **replays**.

Hash-chained journal

```
entry_hash = sha256(ts + kind +  
data + prev_hash)
```

Tamper-evident by construction.
`verify-journal` proves the chain; a
single edited byte breaks it.

Bit-for-bit replay

Every tick stores its full input
snapshot. `tools/replay.py` re-runs
the deterministic pipeline over the
week and diffs each decision. Current
score: **100% MATCH**. Claims: **20/20**,
zero mismatches.

Claims reconciler

`tools/reconcile.py` recomputes
every number in the write-up from
the journal — no credentials
required. Evidence archive snapshots
account/orders/positions daily at the
close.

DEVLOG — 32 documented self-corrections

The build found real things: Alpaca's mleg **limit price is signed** (negative = credit — our `abs()` once turned «collect ≥ 3.40» into «pay ≤ 3.40», caught by a weekend acceptance test, fixed, upstream feedback written) · paper fills options against Friday quotes while closed · battery defaults silently skip scheduled runs · a far-OTM hedge converts the tail, not the worst case. Each one is in the repo with its fix and test.

make verify = chain + replay + reconcile, one command, green

Four books. Same inputs. Different brains.

BOOK	WHAT IT IS	P&L (UPDATED AT SUBMISSION)
real	the full agent: desk + 18 gates + management	+\$763
shadow_nogates	same strategy, gates ignored, fills at mid	+\$61 — the gates are worth \$702
shadow_nohedge	the real book minus its hedge sleeve	+\$772 — the tail cost \$9 so far
baseline_naive	«LLM reads a headline, buys ATM» — what a coin flip earns	-\$46 — negative all week

We didn't just claim the gates and the structure matter — we ran the agent without them, on the same ticks, and published the gap.

Marks carry net book greeks (\$-delta / \$-theta / \$-vega) — the dashboard explains WHY the curve moved

RESULTS — PAPER ACCOUNT PA39C10YAMYQ (FINAL NUMBERS IN THE SUBMISSION)

Small, **explained**, and reproducible P&L beats a lucky coin flip.

The week in one story

Day one: vol was cheap by our own measure — the agent bought convexity instead of selling premium. The market slid; three of four puts closed at **+64...70%** for **+\$764** realized. Then realized vol collapsed to 7.2% against 12.4% implied, the regime flipped **rich**, and the desk sold its first iron condors — the structure it was built for. Both branches, live, on their own signal.

Risk that never left the box

Max structural loss per position $\leq$ 1.25% equity · book worst case gate-capped at 7.5–8% · long-premium sleeve $\leq$ 1.5–2.5% (earned) · NFP-morning deadline entered with winners locked by rule.

Built and operated by AI

Designed, coded, red-teamed, debugged and run end-to-end by Claude (Claude Code + Alpaca MCP), with 32 self-corrections documented in DEVLOG.md — including the desk replaying its own snapshot to find a **\$208** mispriced exit, then pinning it with a test. Disclosed openly.